Biplov Pokhrel

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Education

Technische Universität Chemnitz — Chemnitz, Germany
Masters, Automotive Software Engineering — *Oct 2025 - Present*

- Courses: Hardware Software Codesign, Embedded Systems, Hardware Development with VHDL, Automotive Software Engineering, Design of Software for Embedded Systems, Advanced Platforms for Automotive Systems, Automotive Sensor Systems, Neurocomputing, Realtime Systems

National Institute of Technology, Rourkela — Rourkela, India
B.Tech. in Electronics and Communication Engineering | Grade: 1.6 — *June 2019 -May 2023*

- Thesis and Project: IoT Interface For Real-Time Voltage Mode Pulse Oximetry Using FERN Stack

Experience

Lowe's India — Bangalore, India
Software Engineer — *May 2025 - Oct 2025*

- Owned features end-to-end, from planning and design through planning, implementation, testing, deployment, and production support
- Collaborated with product team to come up with technical solutions to new features
- Generated Architectural Design Resources and guided team members for implementation
- Integrated OpenAI APIs in existing architecture for digitization of handwritten data
- Created migration tools to replace existing Apache NiFi based data migration system with visualization and validation mechanisms
- Deeply involved in feature deployment, bug fixes, production issues resolution and product stability management
- Involved in confluence documentation, PR reviews and suggestions, planning, design, refinement, etc.

Lowe's India — Bangalore, India
Associate Software Engineer — *June 2023 - May 2025*

- Developed and maintained backend microservices using Java, Spring Boot, and Spring WebFlux for enterprise retail applications
- Implemented RESTful APIs and business logic while collaborating with senior engineers on feature development
- Developed and integrated React-based frontend components with backend REST APIs to deliver end-to-end application features
- Collaborated with frontend engineers to integrate React applications with backend microservices and REST APIs
- Built and maintained event-driven integrations using Apache Kafka, Apache Camel, and Apache NiFi
- Developed SQL queries and worked with PostgreSQL and Spring Data JPA for efficient data persistence
- Wrote unit and integration tests, resolved production defects, and participated in code reviews to improve software quality
- Worked in Agile teams using Git, Jenkins, Docker, Jira, and CI/CD pipelines for continuous delivery

Proof and Experimental Establishment Lab, DRDO — Chandipur, India
Intern — *2022*

- Extracted object coordinates from featureless terrain using photogrammetry, image processing, feature extraction, and camera calibration
- Applied machine learning techniques for analysis and optimization
- **Tech stack:** Python, PyTorch, OpenCV, Skimage, NumPy, Matplotlib

Skills

OS – Linux • MacOS • Windows

Languages – Java • C • C++ • Python • JavaScript

Backend – Spring Boot • Spring WebFlux • PostgreSQL • MongoDB

Frontend – React • HTML • CSS • Tailwind • Storybook

Testing – JUnit • Mockito • Cypress • Karate • Jest

Messaging – Apache Kafka • Apache Camel • Apache NiFi • Apache Superset

DevOps – Docker • Kubernetes • Jenkins • Sonar • Snyk • Kibana • Grafana

Tools – Jira • Postman • Swagger/OpenAPI • Git • Postman • RapidAPI

Projects

RASPNet – Multi-Node Ring Network Firmware (AVR/C) May – Jul 2026

- Bare metal programming
- Implemented a 3-layer bit-serial ring network protocol stack in C for ATmega328P, including custom framing, streaming CRC-32 validation, and cut-through packet relaying
- Creted ISR-driven bit-level clock/data synchronization with sliding-window preamble detection for reliable inter-node communication
- Optimized SRAM footprint on a resource-constrained MCU by eliminating static buffers and redesigning ring-buffer memory management
- Built a custom Python-based serial web terminal (SSE streaming, raw termios) for live protocol debugging and multi-node monitoring

IoT Interface for Real-Time Voltage Mode Pulse Oximetry Apr 2023

- Built IoT-enabled pulse oximeter with low-power transimpedance amplifier and real-time SpO2/heart rate monitoring
- Integrated Firebase with React JS for real-time data visualization
- Handled software-hardware integration and high-frequency data acquisition

AI Powered Crowd and Breach Detection System Apr 2022

- Developed AI-powered crowd and perimeter breach detection system using Python, OpenCV, TensorFlow/PyTorch with 92% detection accuracy
- Implemented MERN-based UI with real-time anomaly detection and automated alerts

Other personal projects

- Calorie Tracker
- Expense Tracker
- Task Set Scheduler
- Flashcard Review System
- Phone Controlled Multimedia Interface for TV

Languages

English (C2) **German (B1)**